Social Media Sentiment Analysis at Scale

Image source: Irasutoya

Ewa Herman
Julia Krempińska
Julian Sungkyun Yoo
Natalia Przewdzięk

How did we setup our team?

Communication Tools

Platform for code

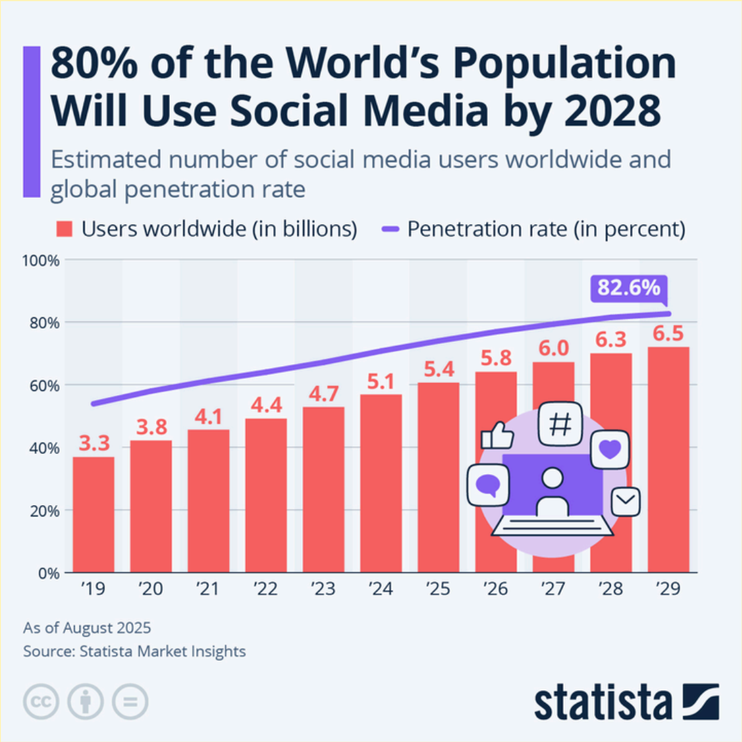
github.com/JKrempinska/machine_learning_project

Roles

Scrum

2-week cadence

Why did we choose this topic?



It is fascinating to observe **how machines can learn the nuances of human communication.**

Can help with

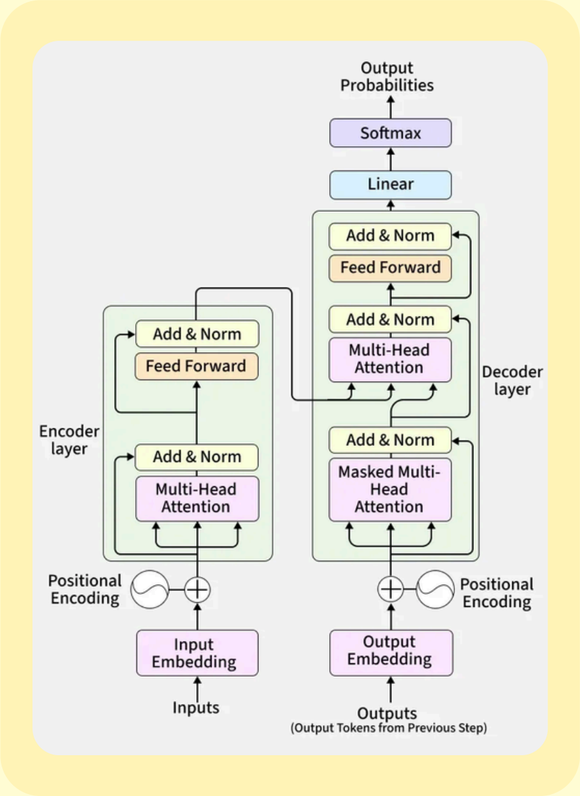
- monitoring brand sentiment to inform **strategic planning**
- **benchmarking brand's perception** against competitors
- evaluating impact of **marketing** campaigns
- researching **market** and identifying **trends**

Core Idea

Potential Approaches

NLTK

NLP + DL



LLMs

Potential Datsets

IMDb

kaggle™

Labels of the sentiment on each post

Potential Constraints

Thank you

Image source: Irasutoya

Reference List

[1] Vaswani, A., et al. Attention is all you need. (2017). Retrived from: https://arxiv.org/pdf/2104.10662

[2] Garcia, D., et al. Social Media Emotion Microscopes Reflect Emotional Experiences in Society at large. (2021). Retrieved from: https://arxiv.org/pdf/2107.13236

[3] Diana, T. Using sentiment analysis to reinforcement learning: The case of airport community engagement. (2022) Retrieved from: https://doi.org/10.1016/j.jairtraman.2022.102228

[4] Rodriguez-Ibanez, M., et al. A review on sentiment analysis from social media platforms. (2023). Retrieved from: https://doi.org/10.1016/j.eswa.2023.119862

[5] Fourreau, V., 80% of the World's Population Will Use Social Media by 2028. (2025). Statistia.